Annex XVI — WR Code®: Transport-Agnostic Reference and Resolution Architecture

Version 1.2 — 5 August 2026

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Normative language. MUST, MUST NOT, SHOULD, SHOULD NOT, and MAY are used as in RFC 2119. Sections marked informative do not constrain conformance.

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§XVI.1 Purpose and Scope

§XVI.1.1 The problem this annex resolves

The designation "WR Code" invites a category error: the assumption that a WR Code is a visual barcode — a QR code with additional branding, payload conventions, or styling. This annex removes that ambiguity normatively.

Non-Identity Clause (normative). The WR Code is not a symbology. It is not defined by, and MUST NOT be equated with, any visual, radio, acoustic, or textual carrier format. A QR code, Data Matrix, AprilTag, NFC tag, BLE beacon, printed character string, or email payload that carries a WR identifier is a carrier instantiation of a WR Code — never its definition.

Canonical definition (normative). A WR Code is a transport-agnostic, machine- or human-readable reference to a verifiable WR entry — a Public WR Handshake, an entry point, or an object — whose meaning, identity, and trustworthiness arise exclusively from resolution and validation within the WR infrastructure, never from the carrier itself.

This definition is the structural companion to the canonical initiation-layer statement of Annex IX (§IX.3.1): the code is an identifier and invitation — not authority. Annex IX governs what a code is not permitted to do; this annex governs what a code is, independently of how it travels.

§XVI.1.2 What distinguishes this architecture (informative)

The surrounding ecosystem of machine-readable entry mechanisms — QR deep links, NFC NDEF records, GS1 Digital Link, app-invocation clips, proximity beacons — shares one behavioral pattern: **identifier → URI → immediate action**. Scanning opens a browser, launches an application, or triggers a platform behavior. The carrier's content is treated as an instruction.

The WR architecture binds the identifier differently: **identifier → repository resolution → publisher validation → offer**. Between recognition and effect stands a mandatory trust and consent chain. The difference is not the carrier and not the encoding; it is that no carrier, in any WR deployment, can shorten this chain. Section §XVI.17 records this distinction against prior work without asserting novelty claims.

§XVI.1.3 Scope of this annex

This annex specifies:

1. the transport-agnostic definition and reference model of the WR Code (§XVI.1, §XVI.3, §XVI.4);
2. the public identifier scheme (§XVI.5);
3. repository resolution, carrier tokens, and publisher determination (§XVI.6, §XVI.7);
4. the entry lifecycle and the permanence guarantees of published identifiers (§XVI.8);
5. carrier profiles and their capability boundaries (§XVI.9);
6. the WR Code Generator as the authoritative emission component for all carrier artifacts (§XVI.10);
7. bootstrap and onboarding semantics (§XVI.11);
8. composite deployments and the Carrier Binding Registry (§XVI.12);
9. session-bound resolution and dynamic sub-entries (§XVI.13);
10. the BLE discovery layer (§XVI.14), offline operation (§XVI.15), and the security model (§XVI.16).

Out of scope are concrete product user interfaces, concrete registry operations tooling, and the visual design language of rendered artifacts beyond the conformance-relevant rendering requirements of §XVI.10.4.

§XVI.1.4 Relationship to other annexes

- **Annex VII —** WR Handshake architecture, capability declaration, publisher attestation. This annex terminates every successful resolution in a WR Handshake offer governed by Annex VII.
- **Annex IX —** initiation layer, invitation classes, code capture methods, channel provenance, the Link-as-Entry failure class, and the WR TED execution model. The concrete code scheme of Annex IX is hereby positioned as one carrier-facing encoding profile of the transport-agnostic model defined here. **Identifier-class boundary (normative):** the Baseline Code of this annex identifies public WR entries of DNS-verified publishers; invitation codes for private or temporary invitations between parties without a public

publisher context remain a distinct identifier class governed by Annex IX. Convergence of the two classes is not required by this annex.

- **Annex X** — WR Guard™ and capability-governed information flow. Resolution and bootstrap traffic crossing trust boundaries fall under Annex X disciplines.
- **Annex XI** — WR Pointer™ and designation instruments. Voice, vision, gaze, and pointer are discovery instruments under Annex XI, not carriers (§XVI.4.2).
- **Annex XIV** — Scoped Worlds. Spatially anchored deployments of carriers inherit Scope semantics.
- **Annex XV** — physical-world object designation, Floor Entry, Spatial Anchor Registry, and Instance Attestation (§XV.19). The compact-field WR Code used for Floor Entry in Annex XV is a conformant carrier profile of this annex (§XVI.6.2, §XVI.9.2).

Nothing in this annex weakens any invariant of the annexes above. Where this annex and an earlier annex describe the same mechanism, the earlier annex remains authoritative for its domain and this annex supplies the cross-carrier generalization.

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§XVI.2 Definitions

WR Entry. A repository-registered referent of a WR Code: a Public WR Handshake, an entry point, or an object identity.

Umbrella Handshake. A publisher-level Public WR Handshake serving as the stable public entry to a publisher's offering, beneath which object-, context-, or action-specific sub-entries can be created and resolved dynamically (§XVI.13).

Carrier. Any medium that transports or exposes a technical identifier referencing a WR Entry: QR, Data Matrix, AprilTag/ArUco, NFC, BLE payload, printed or displayed character string (OCR-capturable), email/BEAP™ payload, dynamic display.

Discovery Instrument. A method that determines which entry or object is meant, without itself transporting an identifier as payload: BLE proximity sensing on the receiving side, vision recognition, gaze, head direction, voice addressing, WR Pointer™, manual entry initiation. Discovery instruments are governed by Annex XI and Annex XV; this annex references them only where they interact with carriers.

Baseline Code. The human-readable public identifier of a WR Entry (§XVI.5), capturable by reading, typing, dictation, telephone relay, or OCR.

Universal Baseline Carrier. The carrier class consisting of the Baseline Code in visible or spoken form. It is the only carrier class that is mandatory for every public WR Entry (§XVI.7).

Augmentation Carrier. Any carrier other than the Universal Baseline Carrier. Augmentation carriers add capability or comfort; they never replace the baseline (§XVI.9).

Carrier Token. An opaque identifier embedded in a carrier that the repository resolves to a WR Entry. A carrier token need not contain the Baseline Code or any human-meaningful component.

Carrier Binding Registry. The per-entry repository record listing all legitimately deployed carriers of that entry, per Deployment Site where applicable (§XVI.12).

Deployment Site. A publisher-declared physical or logical placement context for an entry's carriers — a shop entrance, a shelf, a shop window, a printed label series, an email template, a web page — optionally bound to spatial anchors under Annex XV. The Deployment Site scopes the composite-deployment validation of §XVI.12.

WR Code Generator. The authoritative emission component that produces all carrier artifacts of a WR Entry from its single versioned entry record and registers them in the Carrier Binding Registry (§XVI.10).

Output Profile. A generator configuration selecting which carriers are emitted for an entry and with which rendering parameters (§XVI.10.2).

Bootstrap Payload. A dual-audience carrier payload that a WR-enabled runtime resolves natively and a non-WR device interprets as an informational URL (§XVI.11).

Resolution. The repository-mediated mapping from an identifier or carrier token to a WR Entry, including publisher, status, version/generation, permitted context, and trust material.

Session-Bound Resolution. Resolution of a public identifier performed within an established BEAP™ session with the resolved publisher, attaching the session context so that dynamic sub-entries become resolvable (§XVI.13).

Entry Status. The lifecycle state of a WR Entry: active, inactive, revoked, superseded, or compromised (§XVI.8.1).

Workflow Ready. The canonical meaning of the WR series: a securely described and validated workflow is available as an option. Availability is never execution (§XVI.3, P4).

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§XVI.3 Core Principles

The following principles are normative for every carrier, every discovery instrument, and every deployment profile.

P1 — Transport agnosticism. The same WR Entry MAY be reachable over any number of carriers and discovery instruments. The security and identity model MUST be identical across all of them. No carrier confers, weakens, or substitutes trust.

P2 — Identifier, not authority. A carrier transports or exposes an identifier. It carries no permission, no session, no credential, and no execution right. This restates Annex IX §IX.3.1 across all transports.

P3 — Publisher determinability. In every public deployment, the publisher behind an entry MUST be unambiguously determinable through authenticated resolution — never through an unverified claim, and never left ambiguous. Two conformant determination profiles exist (§XVI.6.3).

P4 — Recognition is not execution. Capturing, recognizing, receiving, or resolving a WR Code MUST NOT execute any workflow, open any external content, or create any authorization. Resolution terminates in a displayed, validated offer. Execution requires the full Annex VII / Annex IX consent and, where applicable, action-bound finalization chain (Annex IX §IX.16 ff., Annex XV §XV.19).

P5 — Universal Baseline Rule. Every public WR Entry MUST be reachable through the Baseline Code alone. No public entry may exist that is reachable exclusively through QR, NFC, BLE, fiducials, or any other augmentation carrier.

P6 — Augmentation, not replacement. Every non-baseline carrier is defined by what it adds to the baseline — payload capacity, pose, proximity, tap semantics, channel provenance — never by replacing it.

P7 — Uniform resolution chain. All entry paths follow the same chain: carrier or discovery → technical identifier → repository resolution → publisher validation → WR Handshake offer → deliberate selection and consent → (where applicable) separate action-bound authorization and execution. No layer may be skipped; no lower layer may trigger a higher one autonomously.

P8 — Fiducials and tokens are candidates, never proof. Optical fiducials, NFC tags, and BLE tokens are copyable. They designate candidates. Effects require the attestation disciplines of Annex XV §XV.19 and the consent classes of Annex IX. A copied carrier must never suffice for impersonation or authorization.

P9 — Proximity is not identity. Physical or radio proximity to a carrier or beacon proves nothing about either party. BLE in particular answers only "what might be near me," never "who may execute what" (§XVI.14).

P10 — Privacy by non-emission. Entry mechanisms MUST NOT turn the user's device into a trackable emitter. The user device does not broadcast a stable public WR user identifier during discovery (§XVI.14.3).

P11 — Identifier permanence. Publisher parts and published local identifiers are assigned exactly once and are never reassigned — not after revocation, not after publisher offboarding, not after expiry (§XVI.8.3). A published identifier permanently denotes the entry it was issued for, or nothing.

P12 — Local re-rendering. A carrier rendering received from an external party — an image in an email, a graphic in a message, a fetched artifact — is a transport artifact only. Runtimes MUST NOT display received renderings; every visual representation of a WR Code shown to the user is generated locally from data that has passed resolution and validation (§XVI.6.5, §XVI.10.5).

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§XVI.4 The Seven-Layer Resolution Model

§XVI.4.1 Layers (normative)

Every WR entry path decomposes into seven layers. Conformant implementations MUST preserve the layer boundaries; a carrier profile declares which layers it serves and which it is prohibited from serving.

1. **Discovery / Selection.** Determines which entry or object could be meant. Instruments: BLE proximity sensing, vision, gaze, head direction, WR Pointer™, voice, manual initiation. Governed by Annex XI/XV.
2. **Carrier / Transport.** Transports an identifier or carrier token: QR, Data Matrix, AprilTag, NFC, BLE payload, email/BEAP™, printed or displayed text (OCR).

3. **Resolution.** The WR repository maps identifier or token to publisher, entry, handshake, object, status, version/generation, and permitted context.
4. **Trust Validation.** Publisher and entitlement are validated: DNS and domain binding, WR Connect™, SSO, email channel authentication (Annex IX §IX.1.9–1.10 disciplines), cryptographic signatures, attestation classes P0/P1/P2 (Annex VII).
5. **WR Handshake.** The validated offer: who offers, which capabilities, which interaction would be possible (Annex VII).
6. **Selection and Consent.** The runtime displays the validated option with publisher, verified domain, offered handshake, possible action, and required consent. The user decides deliberately.
7. **Execution.** Only upon a finalizing action does an action-bound authorization and execution process occur (Annex IX WR TED lifecycle; Annex XV Instance Attestation where a physical instance is involved).

§XVI.4.2 Layer discipline (normative)

1. Layers 1 and 2 MUST NOT trigger layers 5–7. Discovery and transport produce candidates and identifiers, nothing more.
2. Layer 3 output is untrusted until layer 4 completes. A resolved but unvalidated entry MUST NOT be presented as trustworthy.
3. Layer 6 is mandatory for every entry path, including assisted and headless captures (Annex IX Assisted Code Capture; Annex XV Headless WR Code Scan). Consent completes entry; nothing else does.
4. Voice, vision, gaze, and pointer are layer-1 instruments, not layer-2 carriers. An earlier working formulation listing them among carriers is superseded. Where a discovery instrument yields an identifier (e.g., vision reading a printed Baseline Code), the identifier itself enters at layer 2 as an OCR capture; the instrument remains layer 1.
5. A carrier profile MUST declare prohibited layers, and runtimes MUST enforce the prohibition (e.g., BLE: layers 4–7 prohibited, layer 3 only runtime-mediated, §XVI.9.2).

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§XVI.5 The Public Identifier Scheme

§XVI.5.1 Structure (normative)

The Baseline Code is a 12-character alphanumeric identifier:

```
P P P P P P L L L L L C
└─ publisher ─┘└─ local ─┘└ check
      (6)           (5)      (1)
```

- **Publisher part** — 6 characters, fixed length. Identifies the publisher. Assigned within the WR registry.
- **Local part** — 5 characters by default, extensible. Identifies a handshake, entry point, or object within the publisher's namespace. Handshakes and objects share this namespace; the referent type is resolved by the repository, not encoded in the identifier (§XVI.5.6).
- **Check character** — always the final character. Error detection for typing, dictation, telephone relay, and OCR (§XVI.5.4).

§XVI.5.2 Extension rule (normative)

Publishers with exceptionally large customer or object populations MAY use an extended local part. The extension mechanism is:

1. The publisher part is always exactly 6 characters.
2. The check character is always the final character.
3. The local part is everything between them: 5 characters by default, extensible by appending one or more characters (typically one or two).

This makes every code self-delimiting without a length field or version marker: any parser derives the structure from total length alone. The grouped presentation (§XVI.5.5) was chosen precisely to keep this extension visually and cognitively simple.

Capacity (informative): with the 32-symbol alphabet of §XVI.5.3, the publisher part addresses ≈ 1.07×10^9 publishers; the default local part ≈ 3.3×10^7 entries per publisher; each appended character multiplies local capacity by 32. The identifier-permanence rule (P11, §XVI.8.3) consumes address space permanently; at these magnitudes the consumption is immaterial, and the extension rule absorbs any publisher-local exhaustion.

§XVI.5.3 Alphabet (normative)

The alphabet is **Crockford Base32**: the ten digits `0`–`9` and the letters `A`–`Z` excluding `I`, `L`, `O`, and `U` — 32 symbols. Capture is case-insensitive, and capture paths MUST apply the Crockford decoding convention before validation: fold case, map the letters `I`/`i`/`L`/`l` to the digit `1` and `O`/`o` to the digit `0`. The symbol set is fixed for all conformant identifiers; deployment locales MAY add capture-side disambiguation aids (display typography, dictation phrasing) but MUST NOT alter the symbol set.

Rationale (informative): excluding `I`, `L`, and `O` removes the canonical visual confusables (I/1/l, O/0); excluding `U` avoids U/V confusion and accidental obscenities in generated codes. Excluding confusables trades address space for capture reliability; the extension rule (§XVI.5.2) absorbs the capacity cost. Crockford Base32 is an openly published, unencumbered encoding convention with ubiquitous independent implementations.

§XVI.5.4 Check character (normative)

The check character is computed with the **Damm algorithm** over a totally anti-symmetric quasigroup of order 32 aligned to the alphabet of §XVI.5.3, evaluated over the canonical ungrouped form (§XVI.5.5) excluding the check character itself. The Damm construction detects (a) all single-character substitution errors and (b) all adjacent-transposition errors at any conformant length, natively over the variable-length identifier — the required error-detection property of this scheme. The concrete quasigroup table and its test vectors are published as registry material and are identical for every deployment; an identifier generated or validated with a different table or algorithm is non-conformant.

Rationale (informative): simple parity and Luhn-mod-N schemes were rejected because they miss transposition errors; totally anti-symmetric quasigroups exist for every order except 2 and 6, so order 32 is well-defined. The Damm algorithm is an openly published, unencumbered mathematical method.

A failed check MUST be reported as a capture error and MUST NOT trigger resolution. Runtimes SHOULD offer character-level correction assistance on check failure.

Length errors (informative). Because the local part is variable-length (§XVI.5.2), a character omission or insertion produces a candidate that parses under a different structure and is caught by the check character only probabilistically (≈ 31/32 with a 32-symbol alphabet). The residual risk is bounded by the mandatory validated presentation of layer 6: a capture that survives an undetected length error still resolves to a real entry whose publisher, domain, and offer are displayed before any consent, so a mis-capture cannot silently reach a different effect — only a visibly different offer.

§XVI.5.5 Canonical form and display grouping (normative)

The canonical identifier is the ungrouped character sequence. Display, print, and speech MAY group it for readability — reference grouping `PPPPPP-LLLLL-C` (e.g., `ABC123-DEF45-X`) — and vision/OCR capture MUST accept grouped, ungrouped, hyphenated, and spaced presentations, normalizing to canonical form before check verification and resolution. The check character is computed over the canonical form only. Renderings emitted by the WR Code Generator (§XVI.10) use the reference grouping unless an Output Profile specifies otherwise; all groupings of the same canonical sequence are the same identifier.

§XVI.5.6 Type resolution (normative)

The identifier does not encode whether it references a handshake, entry point, or object. The repository resolves the type, and the runtime displays it after validation. *Rationale (informative):* type-in-identifier costs characters, adds a dictation failure mode, and duplicates information the mandatory resolution step supplies authoritatively.

§XVI.5.7 Relationship to internal identifiers (normative)

The Baseline Code MAY be the entry's primary key or MAY be an alias resolving to a larger internal identifier (96/128-bit class). Both models are conformant. In either model, resolution output — not identifier arithmetic — is authoritative for identity, and internal identifiers never surface in user-facing capture paths.

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§XVI.6 Repository Resolution and Carrier Tokens

§XVI.6.1 Full-address carriers and pointer carriers (normative)

Two carrier payload classes exist:

- **Full-address payloads** carry the complete Baseline Code (optionally with supplementary fields). They resolve without deployment context. The Universal Baseline Carrier is always full-address.
- **Pointer payloads** carry an opaque, globally unique Carrier Token (e.g., AprilTag family + tag ID; BLE beacon token; NFC token). The repository maps the token to publisher, entry, status, version/generation, permitted context, and trust material. The full Baseline Code need not be extractable from the carrier.

Baseline carriers MUST be full-address. Augmentation carriers MAY be either class.

§XVI.6.2 Supplementary carrier fields (normative)

Carriers with payload capacity (QR, Data Matrix, NFC, email) MAY carry supplementary fields — manifest identifier, expected manifest hash, code anchor identifier, protocol version, signature — as in the Annex XV Floor Entry field set. Supplementary fields are optimizations (fewer resolution round-trips, offline pre-verification, integrity pinning). They are never a precondition for reachability: the repository MUST supply every datum required for validation given the Baseline Code alone.

§XVI.6.3 Publisher determination profiles (normative)

Principle P3 admits exactly two conformant profiles:

- **Explicit-publisher profile.** The identifier contains the publisher part in clear form (the Baseline Code structure of §XVI.5). Mandatory for the Universal Baseline Carrier, since the baseline must resolve from any context — spoken over a telephone, typed on a foreign device — with no deployment context available.
- **Opaque-token profile.** The carrier carries a globally unique token; the repository resolves it unambiguously to publisher, domain, and entry. Permitted for augmentation carriers, whose deployment context (registered beacon, registered fiducial, registered tag) is itself repository-known.

Both profiles satisfy P3; what P3 prohibits is any deployment in which only a locally scoped number is known and the publisher is asserted rather than resolved.

§XVI.6.4 The email carrier exception (normative)

Email/BEAP™ differs from all anonymous carriers: the publisher is determinable from the communication channel itself — sender address and domain, SPF, DKIM, DMARC, DNS verification, and where present WR Connect™ — under the Channel Provenance Verification and Uniform Surfacing disciplines of Annex IX (v1.9/v1.10). The publisher therefore need not be re-encoded in the WR payload. This is a substitution of determination method, not an exemption from P3; where channel authentication fails, the Annex IX unsuppressible-warning discipline applies, and the Link-as-Entry failure class (Annex IX §IX.1.5) remains the governing critique of URL-first email entry.

Anonymous carriers — QR, NFC, AprilTag, BLE, OCR — have no sender context and MUST satisfy P3 through §XVI.6.3.

Private handshakes between private individuals, where no DNS-verified publisher exists, are out of scope for this annex and require a distinct identity and authentication model (identifier-class boundary, §XVI.1.4).

§XVI.6.5 Verification-gated email capture (normative reference flow)

A Public WR Handshake attaches to an email or newsletter as an embedded Baseline Code — visible text in the message body (for example in a newsletter's header or footer block) — optionally accompanied by supplementary fields (§XVI.6.2) and optionally by an attached carrier rendering such as a QR graphic emitted by the WR Code Generator (§XVI.10.2). Capture proceeds as follows:

1. **Depackaging with provenance verification first.** During depackaging, channel provenance is verified before any WR parsing occurs: SPF, DKIM, and DMARC evaluation and DNS verification of the sender domain against the claimed publisher domain, plus the Discovery Record check, per the Annex IX Channel Provenance Verification discipline (v1.9). A typed Channel Provenance Record is emitted for every depackaged message (Uniform Surfacing, Annex IX v1.10).
2. **Verification gates parsing (normative order).** Only when verification succeeds — sender domain authenticated, the resolved publisher exists in the repository, and sender/publisher alignment holds — does the runtime parse the embedded WR code.
3. **Received renderings are never displayed (normative, per P12).** Any carrier rendering attached to or embedded in the message — QR graphic, combined graphic, styled code image — is treated as a transport artifact exclusively. It is parsed, where applicable, as one more source of the identifier; it is never shown to the user. Within the BEAP™ surface, only the textual code is presented at depackaging time. Where the inbox or offer surface later presents a visual WR Code, that visual is **re-generated locally** by the runtime from the parsed and repository-validated identifier (§XVI.10.5). Ingestion thereby remains text-only secure: no sender-supplied image reaches the user interface, and a divergence between what a received graphic encodes and what it visually suggests is structurally impossible to exploit — the displayed rendering is always derived from validated data.
4. **Surfacing as a validated offer.** The parsed and repository-validated entry is surfaced in the extension as a Connect offer with a WR Connect affordance, displaying publisher, verified domain, and the offered entry (layer 6).
5. **Failure path.** No WR affordance is ever derived from a message whose channel provenance failed. In that case the Annex IX unsuppressible-alert discipline applies to the surfaced message, and no Connect affordance is offered — offer never action, and no "connect anyway."
6. **Consent chain unchanged.** Activating WR Connect is a layer-6 selection; everything downstream follows the standard chain (P7), including any action-bound authorization.
7. **Manual entry on the same surface.** The extension surface additionally provides Baseline Code entry: the user MAY type or dictate the code there instead (Universal Baseline Carrier) — for example when the newsletter was read on a non-WR client and the code is relayed by hand.

The gate order in step 2 is the load-bearing rule of this flow: parsing an identifier out of unauthenticated mail and presenting it — even with a warning — would re-create the Link-as-Entry failure class with a code instead of a link. Verification first, parsing second, offer third.

Forwarding and mailing lists (informative). Forwarding, mailing-list relaying, and image-stripping proxies routinely break DKIM alignment or remove attachments. This is an expected case of the failure path, not an edge case: the forwarded message yields no Connect affordance, and the visible Baseline Code in the message body remains readable and manually enterable on the same surface (step 7). The Universal Baseline Rule (P5) is what makes forwarding non-destructive by design — the entry stays reachable through the code as text, while the authenticated-channel comfort path is simply unavailable for a message whose channel cannot be authenticated.

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§XVI.7 The Universal Baseline Carrier

§XVI.7.1 Sufficiency (normative)

Per P5, the Baseline Code alone MUST suffice to reach every public WR Entry. Conformant capture paths include, at minimum: manual typing, voice dictation, telephone relay followed by manual entry, and OCR of the printed or displayed code (including Assisted Code Capture per Annex IX v1.8).

The baseline is not a degraded fallback; it is the definition of minimum reachability. Its irreplaceable roles (informative): damaged or unreadable machine carriers, devices without supported scanning, accessibility, speech-only contexts, support and recovery flows, and cross-person relay ("read me the code").

§XVI.7.2 Co-presentation (normative)

Wherever a physical augmentation carrier is deployed, the visible Baseline Code SHOULD be presented alongside it. The AprilTag + printed-code pairing — machine-speed recognition and pose from the fiducial, human-legible and dictatable identity from the code, both resolving to the same entry — is the reference pattern for physical placement, not a special case.

Exception: deployments where visible text is infeasible or undesirable (miniature tags, aesthetic constraints) MAY omit co-presentation if the entry remains baseline-reachable through published means (packaging, signage, listing).

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§XVI.8 Entry Lifecycle and Identifier Stability

§XVI.8.1 Status model (normative)

Every WR Entry carries exactly one Entry Status. Resolution output always includes the status, and the runtime behavior per status is normative:

- **active.** The entry resolves normally; validation proceeds; the offer is presented (layer 5–6).
- **inactive.** The entry is temporarily withdrawn by the publisher. Resolution returns publisher, verified domain, and status; the runtime displays the entry as currently not offered and MUST NOT present a handshake offer. Inactive is reversible without identifier change.
- **revoked.** The entry is permanently withdrawn. Resolution returns publisher and status; the runtime MUST NOT present an offer and MUST display the revocation plainly. Revocation is terminal for the entry; the identifier is never reused (§XVI.8.3).
- **superseded.** The entry has been replaced. The repository record carries an explicit successor reference. The runtime displays the supersession — publisher, old entry, successor entry — and MAY offer to resolve the successor. The successor's offer is presented only after the successor's own full resolution and validation. Supersession is always surfaced; silent redirection is prohibited.
- **compromised.** The entry or its deployment is flagged as compromised. Treated as revoked, with an additional explicit warning that the runtime MUST display and MUST NOT allow to be suppressed (following the Annex IX unsuppressible-alert discipline). Where a successor exists, it is surfaced as under superseded, after the warning.

Resolution of a known identifier never fails silently: even terminal states resolve to a status surface. An identifier unknown to the repository is a capture error, not a status.

§XVI.8.2 Permanence by default; expiry as the exception (normative)

Public WR Entries of verified publishers are **permanent by default**: they carry no expiry and remain active until the publisher transitions them per §XVI.8.1.

An optional `expires_at` MAY be set for temporary entries — invitations, sessions, one-time actions, short-lived campaigns, dynamic sub-entries (§XVI.13). On expiry, the entry transitions automatically to **revoked** (default, for one-time semantics) or to **inactive** (publisher-configured, for re-activatable entries). Expiry MUST NOT be imposed as a blanket requirement on WR Codes; mandatory rotation of permanent public identifiers is explicitly rejected as a conformance requirement.

§XVI.8.3 Identifier permanence (normative, restates P11)

1. **Publisher parts are assigned exactly once.** A publisher part, once assigned, is never reassigned — not after publisher offboarding, dissolution, or registry-initiated removal. A retired publisher part permanently resolves to the retired publisher's terminal status.
2. **Published local identifiers are assigned exactly once.** Within a publisher namespace, a local part that has ever been published — including revoked, superseded, and expired entries — is never reassigned to a different referent.
3. No quarantine or cool-down model is used; permanence is unconditional. *Rationale (informative):* printed and physically deployed codes have unbounded lifetimes. Any reassignment — however delayed — would allow an untouched, genuine artifact to resolve to a different publisher or referent than the one it was issued for, defeating P3 without any attacker action. Against an address space of $\approx 1.07 \times 10^9$ publisher parts and per-publisher local spaces extensible without limit (§XVI.5.2), the permanent consumption of identifiers, including identifiers that were never or only briefly used, is immaterial.

§XVI.8.4 Stable identifier, rotatable trust material (normative)

The published identifier and the entry's security material have deliberately different lifecycles:

1. The **Baseline Code and all registered Carrier Tokens of an entry remain stable** across: cryptographic key rotation, signature renewal, certificate and attestation renewal, manifest updates, domain re-verification, resolution-infrastructure changes, and generation increments of the entry record.
2. **Trust material rotates server-side.** Keys, signatures, tokens internal to resolution, manifests, and bindings MAY be renewed at any time. Resolution output carries the current generation; cached material older than the current generation demotes to unvalidated per §XVI.15.
3. **Rotation never requires re-emission.** No key or binding rotation obliges the publisher to reprint, re-tag, re-flash, or re-send any deployed carrier artifact. This property is what makes permanent physical deployment economically and operationally viable (§XVI.10.4).
4. **Supersession is the only identifier-level rotation.** A new identifier is issued only on deliberate publisher decision, organizational migration that changes the referent, or identifier-level compromise (e.g., a mispublished code) — via the superseded status with

explicit successor reference (§XVI.8.1). Cryptographic events alone never force supersession.

Rotation triggers (informative): compromise of trust material, mispublication, organizational migration, migration of cryptographic schemes, change of an underlying binding, or deliberate publisher decision — the first five rotate trust material under a stable identifier; only the cases in rule 4 touch the identifier itself.

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§XVI.9 Carrier Profiles

§XVI.9.1 Profile obligations (normative)

Each carrier profile declares: (a) payload class (§XVI.6.1), (b) layers served and layers prohibited (§XVI.4), (c) capabilities added over the baseline, (d) carrier-specific security and privacy requirements. Runtimes MUST enforce the prohibitions.

Further optical 1D/2D symbologies (e.g., Aztec, classical linear barcodes, additional fiducial families) enter as profiles under exactly these obligations; no symbology holds or acquires special status, and a future purpose-designed native WR symbology would enter as one further Augmentation Carrier profile like any other. Ultrasound/acoustic tokens, UWB, visible-light communication, and future radio classes enter through the same profile obligations.

§XVI.9.2 Profiles (normative unless marked informative)

QR / Data Matrix. Full-address or pointer. Layers: 2 (and bootstrap presentation, §XVI.11). Adds: payload capacity for supplementary fields; fast optical capture at distance; the Bootstrap Payload network effect. Prohibited: 4–7.

AprilTag / ArUco (fiducials). Pointer. Layers: 2; additionally serves layer-1 spatial functions (pose anchor). Adds: machine-speed detection, perspective correction, multi-tag detection, tracking, pose for Floor registration (Annex XV). Prohibited: 4–7. Security: fiducials are copyable; they enter only as candidates and spatial anchors bound through the signed mapping of Annex XV (tag family + ID + size + floor pose + WR identity + replica reference); effects require Instance Attestation.

NFC. Full-address or pointer. Layers: 2. Adds: deliberate tap semantics — precise single-object selection at the shelf, product, or price-tag level within an active context. Prohibited: 4–7. NFC selection is not pairing (Annex XV) and never substitutes a handshake.

BLE. Pointer only. Layers: 1–2 (discovery and token transport). Layer 3 MAY be performed only as runtime-mediated repository resolution; direct on-device resolution outside the runtime is non-conformant. Prohibited: 4–7 absolutely. Adds: ambient proximity discovery with no capture gesture — the carrier of the walk-by scenario (§XVI.14). BLE never resolves to authority, never auto-loads a handshake into the consent layer without user selection, and is subject to the privacy regime of §XVI.14.3.

Email / BEAP™. Full-address. Layers: 2, with layer-4 publisher determination supplied by channel provenance (§XVI.6.4) and capture gated by verification per §XVI.6.5. Adds: authenticated sender context; sealed-relay delivery; Assisted Code Capture from depackaged mail. Received renderings are never displayed (P12, §XVI.6.5.3). Prohibited: 5–7 without the full Annex IX offer/consent path ("offer never action").

Printed / displayed text (OCR, manual, voice). The Universal Baseline Carrier. Full-address by definition. Layers: 2. Adds: universality, accessibility, relayability. Prohibited: 4–7. Mandatory per P5.

Dynamic displays. Full-address or pointer; MAY rotate pointer tokens and MAY present session-bound sub-entry codes (§XVI.13). Layers: 2. Adds: revocability and short-lived deployment-site tokens. Prohibited: 4–7.

Discovery instruments (voice addressing, vision recognition, gaze, head direction, WR Pointer™). Not carriers. Layer 1 only, governed by Annex XI §XI.20 (as amended) and Annex XV. Their output is a designation that either yields a layer-2 identifier (vision reading a Baseline Code) or a repository-known target (Headless WR Code Scan within an established handshake).

§XVI.9.3 Carrier Capability Matrix (normative summary)

Carrier	Payload class	Universal Baseline Carrier	Discovery (L1)	Transport (L2)	Trust (L4)	Execution (L7)	Adds over baseline
Printed/ spoken code (OCR/manual/voice)	full-address	**yes (mandatory)**	—	yes	never	never	universality, accessibility
QR / Data Matrix	full-address or pointer	no	—	yes	never	never	payload capacity, bootstrap
AprilTag / ArUco	pointer	no	spatial	yes	never	never	pose, tracking, multi-detect
NFC	either	no	—	yes	never	never	tap-precise selection
BLE	pointer	no	yes	yes	never	never	ambient walk-by discovery
Email / BEAP™	full-address	no	—	yes	via channel provenance	never	authenticated sender context
Dynamic display	either	no	—	yes	never	never	rotation, revocability, session codes
Voice/ vision/ gaze/ pointer	— (instrument)	n/a	yes	no	never	never	designation without capture

"Never" cells are normative prohibitions. The "Universal Baseline Carrier" column states whether the row **is** the mandatory baseline carrier class — not whether the Baseline Code can appear on that carrier (it can appear on any full-address carrier). The matrix is the tabular form of the Non-Identity Clause: no carrier row reaches trust or execution; those layers belong exclusively to the resolution architecture.

§XVI.9.4 Carrier selection guidance (informative)

Carriers are not interchangeable in properties — only in trust semantics. The security and identity model is identical across all of them (P1); their operational fitness differs by context, and deployments are expected to choose and combine accordingly:

- **Distance vs. reach.** QR and Data Matrix capture optically at distance; NFC selects precisely within arm's reach; fiducials detect at machine speed across a field of view.
- **Gesture vs. ambience.** Scanning and tapping are deliberate gestures; BLE removes the gesture and provides ambient discovery — which is precisely why BLE is confined to layers 1–2.
- **Pose and space.** Only fiducials contribute pose, tracking, and multi-detection; they are the natural choice wherever spatial registration (Annex XV) matters.
- **Onboarding.** Only the QR Bootstrap Payload (§XVI.11) reaches non-WR devices; deployments that expect first-contact audiences benefit from it.
- **Relayability and accessibility.** Only the Baseline Code can be read aloud, typed, dictated, or relayed by telephone — the reason it is mandatory, not optional.
- **Channel authentication.** Only email arrives over an authenticatable channel; it is the one carrier whose publisher determination can ride on provenance (§XVI.6.4).
- **Revocability of the artifact.** Printed carriers are immutable once deployed (their *resolution* stays updatable per §XVI.8.4); dynamic displays can rotate and withdraw the artifact itself.

The following are **conceptual deployment patterns** — disclosed as design options, none of them describing an implemented or existing deployment: a shop window would combine BLE (ambient) with QR (onboarding) and a printed code (relay); a shelf would combine NFC (item precision) with printed codes (fallback); an entrance would combine a fiducial (pose) with QR-or-NFC (capture) — the composite catalogue of §XVI.12.5 enumerates these patterns on the same conceptual basis. The general rule: choose carriers by the capabilities they add (P6); trust never enters the choice, because no choice changes it.

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§XVI.10 The WR Code Generator

§XVI.10.1 Role (normative)

The **WR Code Generator** is the authoritative emission component for all carrier artifacts of a WR Entry. It is a logically single component (its deployment MAY be distributed) with three defining obligations:

1. **One record, many renderings.** The generator produces every artifact of an entry — baseline text, optical codes, fiducial assignments, NFC provisioning records, BLE beacon configurations, print packages, email packages, web packages — from the entry's **single versioned repository record**. It MUST NOT create a separate identity, separate entry, or separate handshake per carrier; every emitted artifact is traceably an instantiation of the same logical entry.
2. **Emission is registration.** Every artifact the generator emits is registered, at emission time, in the entry's Carrier Binding Registry (§XVI.12.1) — carrier type, identifier or token, payload class, and where applicable Deployment Site and pose. The generator is the write path of the registry; the runtime's composite-deployment validation (§XVI.12.2) is its read

path. A carrier artifact that was never emitted and registered is, by construction, an unregistered carrier at validation time.
3. **No secrets in artifacts.** Generator outputs contain only the public identifier or an opaque carrier token, plus optional supplementary fields (§XVI.6.2). Private keys, token-to-entry mappings, registry records, resolution data, and trust material remain exclusively server-side within the repository trust domain.

§XVI.10.2 Output Profiles (normative)

An **Output Profile** is a generator configuration selecting which carriers are emitted for an entry and with which rendering parameters. Output profiles compose from the carrier profiles of §XVI.9; the disclosed reference profiles are:

- **Baseline-only.** The Baseline Code as text ("WR Code only") — for listings, signage copy, support scripts, voice contexts.
- **Baseline + QR.** The reference general-purpose visual profile, optionally as Bootstrap Payload (§XVI.11).
- **Baseline + AprilTag.** Fiducial assignment (family + tag ID from the publisher's allocated range) with registered size and, where deployed spatially, pose — plus the visible code per §XVI.7.2.
- **Baseline + NFC.** An NFC provisioning record (NDEF content or opaque token) for tag writing, with write-protection guidance.
- **Combined physical profiles.** AprilTag + NFC; AprilTag + QR + NFC; BLE + any of the above — emitted as one registered composite for one Deployment Site (the Annex XV combined entrance capture is one such profile).
- **Print package.** PNG, WebP, SVG, and print-ready PDF layouts for labels, signs, and object markings, conforming to §XVI.10.4.
- **Email package.** The elements a publisher embeds in mail: the visible Baseline Code as copyable text, an HTML snippet for body embedding, optionally an attached QR graphic, and the machine-readable BEAP™ structure — such that a publisher sets up a WR-capable mailing without hand-building cryptographic artifacts (§XVI.6.5).
- **Web package.** SVG or web component with visible Baseline Code, structured metadata, and the material for WR Connect™-verified display on the publisher's DNS-verified domain.

Additional output profiles enter without constraint provided obligations §XVI.10.1(1–3) hold. Publishers obtain output profiles through registry tooling, a publisher API, or platform plugins; the integration bar is deliberately low — no publisher-side cryptographic authoring is ever required.

§XVI.10.3 Generator and repository (normative)

Generator and repository are **distinct roles** that MAY be implemented in one service: the repository is authoritative for resolution (read), the generator for emission (write). Both operate on the same entry record; neither introduces state the other cannot see. Carrier retirement — deactivating a beacon, withdrawing a fiducial assignment, ending an email template — is a registry update through the generator role and never destroys the logical entry (P1: carriers can be added, deactivated, or replaced without touching the entry or its other carriers).

§XVI.10.4 Rendering requirements (normative requirement, profiled parameters)

Emitted visual and physical artifacts MUST be reliably capturable under their intended deployment conditions. The conformance-relevant requirement classes are: minimum module and character sizes, quiet zones, contrast, print tolerances, scaling behavior, and the placement and legibility of the visible Baseline Code in co-presented artifacts (§XVI.7.2). The concrete parameter sets are **registry-published rendering profiles** per artifact class; this annex fixes the obligation, not the numbers.

Because resolution stays server-side updatable under a stable identifier (§XVI.8.4), a deployed print artifact never requires re-production for key rotation, offer changes, or handshake evolution — physical re-production is needed only for supersession (§XVI.8.1) or physical wear.

§XVI.10.5 Local re-rendering surface (normative)

The generator's rendering capability exists in two places: centrally, for publisher emission, and **locally in the runtime**, for display. Per P12, whenever a runtime surface shows a visual WR Code that arrived from an external party — an email attachment, a message, a fetched artifact — the displayed visual is produced by the runtime's local rendering from the validated identifier, never by displaying the received artifact. Received renderings are parse-only inputs. This rule makes ingestion text-only secure end to end and structurally eliminates the class of attacks in which a received graphic's encoded content and its visual suggestion diverge (§XVI.16).

Display-to-device handoff (normative). The locally re-rendered visual is a first-class capture surface, not decoration. A second device MAY capture the re-rendered code from the runtime surface — the reference case being a smartphone scanning the code displayed in the desktop inbox or offer surface. The capturing device resolves the same identifier through its own full chain (layers 3–6; session-bound where that device holds a session with the resolved publisher, §XVI.13). The re-rendered code thereby becomes the bridge for cross-device interaction: after its own validated entry, the handheld device can serve as an interaction instrument for the resolved context — canonically as a **voice input surface**, letting the user speak with the resolved context through the smartphone while the primary surface displays it. Because the displayed code is locally generated from validated data (P12), the handoff bridge is by construction as trustworthy as the surface showing it; a received, sender-supplied graphic could never be granted this role. The handoff shortens nothing: the capturing device earns its participation through resolution, validation, and layer-6 consent like any other capture, and voice on the second device remains a layer-1 / interaction instrument (§XVI.4.2.4) — never a carrier of authority.

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§XVI.11 Bootstrap Payload and Network Effect

§XVI.11.1 Dual-audience payload (normative)

A QR (or Data Matrix) deployment MAY encode its payload as a **Bootstrap Payload**: a URL that syntactically embeds the canonical Baseline Code (reference form: `https://<bootstrap-domain>/wr/<canonical-code>`).

- A WR-enabled runtime MUST recognize the bootstrap pattern, extract the embedded identifier, and resolve it natively through the repository (layers 3–6). The URL MUST NOT be dereferenced in this path.
- A non-WR device treats the payload as an ordinary URL and reaches an informational landing page.

This gives every physically deployed WR Code a second function at zero publisher cost: an onboarding channel. A curious passer-by scanning with a stock camera app learns what a WR Code is, who the publisher is, and where to obtain the runtime — the network effect of the QR ecosystem is inherited without inheriting its identifier→action semantics.

§XVI.11.2 Landing page discipline (normative)

The landing page is strictly informational. It MUST NOT execute or initiate any workflow, request credentials, or emulate a handshake or consent surface. It SHOULD display: what a WR Code is, the resolved publisher and verified domain, the nature of the offered entry, and runtime acquisition. After runtime installation, entry proceeds by capturing the same code through the runtime — the URL never becomes the trust anchor. This keeps the bootstrap path outside the Link-as-Entry failure class (Annex IX §IX.1.5): the link explains; the code identifies; the resolution architecture validates.

§XVI.11.3 Bootstrap domain profiles (normative)

Two conformant profiles; deployments MAY offer both.

- **Publisher-domain profile (preferred).** The bootstrap domain is a path or subdomain under the publisher's own DNS-verified domain (e.g., `example-restaurant.tld/wr/ABC123DEF45X`). Preferred because it scales with the publisher population, keeps the publisher — not the infrastructure — in front of the user, and turns the URL into an additional consistency signal: after resolution, the runtime MUST cross-check the bootstrap domain against the entry's verified publisher domain; a mismatch is surfaced as an alert and blocks the trusted presentation of the entry.
- **Central-resolver profile.** A registry-operated bootstrap domain serves publishers without web infrastructure (small shops, market stalls, individuals). The landing page identifies the resolved publisher prominently so the central domain never masquerades as the offering party.

In both profiles the identifier remains authoritative and the URL remains a shell: resolution and validation run against the repository, never against page content.

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§XVI.12 Composite Carrier Deployments and the Carrier Binding Registry

§XVI.12.1 Carrier Binding Registry (normative)

Every WR Entry's repository record includes a **Carrier Binding Registry**: the authoritative list of all carriers legitimately deployed for that entry — per carrier: type, identifier or token, payload class, the Deployment Site where applicable, and where applicable the registered pose or location (feeding, for spatial deployments, the Annex XV Spatial Anchor Registry). The registry is written exclusively through the generator role at emission and retirement (§XVI.10.1(2), §XVI.10.3).

§XVI.12.2 Deployment Sites and validation scope (normative)

The composite-deployment discipline of §XVI.12.3 operates **per Deployment Site**, and its scope is defined precisely to avoid false alarms in dense environments:

1. A **Deployment Site** is a publisher-declared placement context registered for the entry (§XVI.2) — optionally bound to Annex XV spatial anchors for physically registered deployments.
2. The registry check of §XVI.12.3 applies to carriers that **resolve to the same entry**, or that are **presented as members of the same site** — co-presented within one deployment frame, one registered composite artifact (§XVI.10.2), or one spatially anchored region.
3. A physically nearby carrier that resolves to a **different entry of a different publisher** — the neighboring shop's code in a shopping street, an adjacent poster — is **not** an unregistered carrier of this site. It is evaluated independently against its own entry's registry. Proximity alone never places a carrier inside another entry's validation scope.
4. Where a device encounters carriers before any site context exists (ambient BLE, glasses detecting several fiducials), resolution runs per carrier token → entry; site membership is then established from the resolved entries' registered sites, not guessed from physical adjacency.

§XVI.12.3 Baseline-first validation discipline (normative)

Within a composite deployment, the Baseline Code is the trust root of the Deployment Site. Companion carriers are validated against the baseline's repository entry, never the reverse:

1. Resolution of any carrier of the site MUST terminate in the same WR Entry.
2. A carrier presented as a member of the site (§XVI.12.2.2) that is not listed in the entry's Carrier Binding Registry MUST NOT be resolved as trusted. The runtime MUST surface it as an **unregistered carrier** — a warning, never a silent redirection.
3. Where the device encounters a companion carrier first (smart glasses detecting a fiducial or beacon before anyone reads the code), resolution runs token → entry; the entry names the Baseline Code, which the runtime MAY display for human confirmation. Both directions terminate in the same entry; the registry makes the binding verifiable in both.
4. Divergent resolution among carriers of one Deployment Site is an alarm condition, never a feature. The runtime MUST refuse the trusted presentation and report the divergence.

§XVI.12.4 Threat scoping (normative)

The registry check and the no-authority principle address two distinct attack classes, and the annex claims exactly this:

- **Injection — prevented.** An attacker affixing a foreign carrier (rogue QR sticker, extra fiducial, hostile beacon) inside a genuine Deployment Site cannot redirect users silently: the foreign carrier is absent from the site's Carrier Binding Registry and is refused as trusted (§XVI.12.3.2).
- **Cloning — neutralized, not prevented.** A copied genuine carrier placed elsewhere carries identical data and resolves identically. The clone therefore reveals only what is public and confers nothing: carriers are candidates, never authority (P8), and every effect requires the consent chain and, for physical instances, Instance Attestation (Annex XV §XV.19). Within

spatially registered deployments, a cloned fiducial at the wrong location is additionally detectable through its registered pose in the Spatial Anchor Registry.

Formulation for reuse (normative summary): registration prevents injection; the no-authority principle devalues clones.

§XVI.12.5 Composite deployment catalogue (informative, explicitly non-exhaustive)

Any subset of conformant carriers MAY be co-deployed against one entry. The following combinations are disclosed as deployment options; the list is illustrative and deliberately broad, and absence from it implies nothing:

- **AprilTag + printed Baseline Code** — pose and machine-speed capture plus human-legible, dictatable identity (reference physical pairing, §XVI.7.2).
- **QR + AprilTag** — bootstrap/network effect and payload capacity plus precise spatial positioning; the fiducial anchors pose while the QR onboards non-WR devices.
- **BLE + AprilTag** — ambient walk-by discovery plus in-view selection for smart-glasses deployments.
- **BLE + QR (shop window)** — glasses-comfortable proximity discovery plus smartphone onboarding for passers-by.
- **NFC + printed Baseline Code** — tap-precise selection plus read-aloud relayability at shelf or price-tag level.
- **NFC + AprilTag** — tap selection plus pose anchoring at the entrance (the Annex XV combined entrance capture, NFC variant).
- **QR + NFC** — optical capture at distance plus tap capture in reach.
- **Email/BEAP™ + QR in printed attachment** — authenticated channel delivery plus physical re-entry from print.
- **Dynamic display + AprilTag** — rotating short-lived tokens plus stable spatial anchor.
- **BLE + NFC + printed code (retail shelf)** — aisle-level discovery, item-level tap, universal fallback.
- **Combined entrance capture: QR-or-NFC + AprilTag in one frame** — Floor Entry resolution plus pose anchor (Annex XV §XV.1.5 worked example).
- Further combinations of any carriers profiled in §XVI.9, including with discovery instruments of Annex XI/XV layered above them.

The single normative rule behind the entire catalogue is §XVI.12.3: one site, one entry, registered carriers, baseline-first.

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§XVI.13 Session-Bound Resolution and Dynamic Sub-Entries

§XVI.13.1 Resolution contexts and entry granularity (normative)

An Umbrella Handshake is not tied to a single code. A publisher MAY publish **any number of distinct entry WR Codes** within its namespace (§XVI.5.1) that all belong to the same Umbrella Handshake — per location, entrance, shelf, table, product line, campaign, or purpose — and each such code resolves **its specific context immediately**: one capture, one resolution step, and the validated offer already carries the concrete entry context under the umbrella. Pre-published context entries of this kind are ordinary public WR Entries with the full lifecycle of §XVI.8; reaching a concrete context this way requires no session and no second step.

Against this backdrop, a public identifier resolves in one of two contexts:

- **Context-free resolution.** The default path of this annex: any capture, from any surface, resolves through layers 3–6 to the entry's validated public offer. For a publisher-level code this terminates in the **Umbrella Handshake** itself — the publisher-level offer; for a context-specific entry code, the same single step terminates in that entry's concrete context, presented under the same umbrella and the same publisher validation.
- **Session-Bound Resolution.** Where the capturing runtime already holds an **established BEAP™ session** with the publisher that the identifier resolves to, resolution MAY attach the session context. The repository then resolves the same identifier **into the session scope**: in addition to (never instead of) the public referent, the resolution output MAY include or make resolvable the dynamic sub-entries available within that verified relationship. Session binding serves the contexts that are *not* pre-published as entry codes of their own — sub-entries generated dynamically within the relationship (§XVI.13.2).

Attachment is a property of resolution context (layers 3–4), never of the carrier: no carrier can demand, encode, or force session attachment (P2), and the same code captured without a session simply resolves context-free.

§XVI.13.2 Dynamic sub-entries (normative)

Beneath an Umbrella Handshake, a publisher MAY create **dynamic sub-entries**: object-bound, context-bound, or action-bound entries generated on demand within an established relationship — a concrete product offer, a situational workflow, a specific action handshake.

1. **The public entry stays small.** The public identifier need not carry or pre-declare any object, workflow, or action data. It resolves the publisher and the umbrella offer; concrete objects and actions are loaded afterwards inside the verified relationship. This yields context optimization, minimal initial payloads, and situational offers without ever widening the public surface.
2. **Sub-entry identifiers are session-scoped.** Dynamic sub-entries are referenced by session-scoped tokens or short-lived codes (presentable, e.g., on dynamic displays, §XVI.9.2). They are not published Baseline Codes, MUST NOT be relied upon outside the session that created them, and expire with the session or their `expires_at` (§XVI.8.2). Where a publisher promotes a dynamic offer to a durable public entry, it receives a regular Baseline Code under the full lifecycle of §XVI.8.
3. **Publisher identity is inherited, authority is not.** A dynamic sub-entry inherits the publisher validation of the session in which it resolves — the publisher is already authenticated. It inherits **no** consent and **no** execution right: each sub-entry offer passes layer 6 individually, and each action-bound execution requires its own finalization under the Annex IX WR TED lifecycle.

§XVI.13.3 Chain discipline (normative)

Session-Bound Resolution changes **in which context layers 3–4 resolve — never how many layers there are**:

1. The seven-layer chain (P7) applies in full to every sub-entry. Session attachment shortens nothing: no sub-entry skips validation, offer, or consent.
2. Layer-6 consent is per offer. A standing session consent never blankets future dynamic sub-entries.
3. Session-bound resolution output MUST be marked as such, and the runtime presentation distinguishes the session-scoped offer from the public umbrella offer.
4. Loss or expiry of the session removes resolvability of its sub-entries; the public identifier continues to resolve context-free, unchanged.

Relation to earlier annexes (informative): this section generalizes the pattern already present in Annex XV (object resolution after Floor Entry within an established context) and aligns with Annex VII's separation of publisher-level and action-level handshakes; it adds the transport-agnostic statement that any carrier of §XVI.9 can deliver an identifier into either resolution context without gaining or losing any authority thereby.

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§XVI.14 BLE Discovery Layer

§XVI.14.1 Role (normative)

BLE is a discovery and proximity layer exclusively. It is not a trust anchor and not an execution protocol. A shop, shop window, area, or physical object MAY emit a BLE Carrier Token; the repository resolves it to verified publisher, site or location scope, available public handshakes, objects or offers, and where applicable spatial context (as a Scoped-Worlds spatial scope, Annex XIV).

§XVI.14.2 Walk-by pattern (informative reference flow)

1. A beacon signals proximity to a WR-capable site — a shop in a shopping street, a restaurant on the way past.
2. The runtime resolves the beacon token (layer 3, runtime-mediated per §XVI.9.2) and validates publisher and site (layer 4).
3. Matching handshakes or offers appear as selectable options — on smart glasses, without any capture gesture; on a smartphone, as a passive notification surface subject to user-controlled discovery settings.
4. Vision, a fiducial, or NFC subsequently determines which concrete product or object is regarded (layer 1 refinement, Annex XV Sight-Field disciplines).
5. Only upon deliberate selection is the corresponding handshake loaded into the consent surface (layer 6).

Scanning remains available at every step; BLE removes the gesture, not the choice. With smart glasses the walk-by pattern is where ambient discovery becomes genuinely comfortable — which is precisely why the prohibitions of §XVI.14.4 are load-bearing.

§XVI.14.3 Privacy regime (normative)

1. The site emits the discovery token; the user device MUST NOT broadcast a stable public WR user identifier during discovery.
2. Beacon Carrier Tokens SHOULD rotate or be time-bounded; rotation cadence is a registry-published deployment parameter.

3. Resolution occurs under runtime/repository control; raw beacon observations are processed locally and are not reported to publishers.
4. Publishers MUST NOT be able to use BLE deployments to track individual users across sites or over time; conformance of deployment tooling is assessed against this outcome.
5. Proximity is not identity (P9); no presence log, movement history, or dwell profile is created as a side effect of discovery.

§XVI.14.4 Prohibitions (normative)

BLE MUST NOT auto-execute a handshake, MUST NOT start any action without user decision, and MUST NOT create any entitlement. A later action requires the normal handshake, consent, and where applicable action-binding process in full. BLE answers "what might be near me" — never "who may execute what."

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§XVI.15 Offline and Degraded Operation

1. The Baseline Code remains capturable with no connectivity: capture, normalization, and check verification (§XVI.5.4) are local operations. Resolution completes when connectivity returns; the runtime queues the capture and marks it unresolved, never presenting an unresolved capture as validated (layer discipline §XVI.4.2.2).
2. Carriers with supplementary fields (§XVI.6.2) MAY enable offline pre-verification against cached, signed repository material (manifest hash pinning as in Annex XV). Pre-verification narrows uncertainty; it does not substitute layer 4.
3. Cached entries and trust material carry validity bounds and the current generation from the repository (§XVI.8.4); expired cache or a superseded generation demotes the presentation to unvalidated.
4. Extended offline resolution and validation profiles, including trust-material distribution, MAY be defined as implementation profiles; they MUST NOT weaken the layer discipline of §XVI.4.2.

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§XVI.16 Security Considerations (summary)

- **Category confusion** is addressed normatively by the Non-Identity Clause and the capability matrix: no carrier reaches trust or execution.
- **Rogue-carrier injection** is prevented by the Carrier Binding Registry within precisely scoped Deployment Sites (§XVI.12.2–12.4); the site scoping simultaneously prevents false-alarm degradation in dense environments, which would train users to ignore warnings.
- **Carrier cloning** is devalued by P8, the consent chain, Instance Attestation, and optionally detected via spatial binding (§XVI.12.4).
- **Publisher spoofing** is excluded by P3: publisher identity is resolved and validated, never asserted; the bootstrap cross-check (§XVI.11.3) extends this to the onboarding path.
- **Stale-referent capture** — old printed material resolving to a wrong or reassigned referent — is structurally impossible under identifier permanence (P11, §XVI.8.3): a published identifier denotes its issued entry or a terminal status, never a new referent.
- **Rendering divergence attacks** — a received graphic whose encoded payload and visual suggestion differ, or a manipulated email graphic — are structurally eliminated by P12:

received renderings are never displayed; every shown visual is locally re-generated from validated data (§XVI.6.5.3, §XVI.10.5).

- **Link-based phishing** on the email carrier remains governed by Annex IX (Link-as-Entry failure class, Channel Provenance Verification, Uniform Surfacing); the bootstrap landing page is kept outside that failure class by §XVI.11.2.
- **Enumeration and harvesting.** The local-part space is finite and a public resolution interface is, by design, publicly queryable. Local parts MUST be assigned non-sequentially, and public resolution endpoints MUST apply anti-enumeration measures (rate limiting or equivalent cost imposition) as a deployment requirement. The bound on impact is architectural: resolution reveals only data the publisher has made public; bulk enumeration therefore yields a catalogue of public entries, never credentials, sessions, or authority.
- **Resolution metadata privacy.** P10 governs device-side emission; its repository-side counterpart is normative here: repository operators MUST NOT build or expose per-user resolution profiles (who resolved which entry when) beyond what operation strictly requires, and resolution telemetry MUST NOT be made available to publishers at individual-user granularity. Stronger privacy-preserving resolution techniques MAY enter as implementation profiles.
- **Repository availability.** The repository is logically central and topologically distributable: caching hierarchies, replication, and distribution are implementation profiles and MUST NOT alter resolution semantics or the validation chain.
- **Capture-error resolution** (mistyped or misread codes reaching the wrong publisher) is bounded by the check-character requirement (§XVI.5.4, including the length-error analysis) and by mandatory validated presentation before any consent.
- **Consent-surface spoofing** by landing pages or carriers is prohibited (§XVI.11.2) and structurally moot: consent is only meaningful inside the runtime's layer-6 surface.
- **Tracking via discovery** is excluded by the §XVI.14.3 regime and P10.

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§XVI.17 Prior Work and Distinction (informative)

This section records distinctions for defensive-disclosure purposes and asserts no novelty claims.

- **QR deep links / URL-payload codes.** Identifier → URI → action: scanning dereferences. The WR model never dereferences in the trusted path; the Bootstrap Payload confines URL semantics to unenrolled devices and an informational page (§XVI.11).
- **GS1 Digital Link.** Standardizes product identifiers as web URIs with resolver infrastructure; resolution targets are web resources, and trust derives from the URI/domain. The WR model resolves to a validated handshake offer under a mandatory consent layer, with publisher validation independent of the carrier's URI.
- **NFC NDEF / tap-to-launch.** Tap semantics trigger platform actions (open URL, launch app). Under this annex NFC is selection only; no tap reaches layers 4–7.
- **Platform app-invocation mechanisms (app clips / instant-app-class).** Scanning or tapping launches scoped application code. The WR model launches nothing; it produces a validated offer.
- **iBeacon / Eddystone-class proximity beacons.** Broadcast identifiers commonly used for notification and engagement triggering, historically entangled with tracking concerns. The

WR BLE profile is discovery-only under a normative non-tracking regime (§XVI.14.3) and can trigger no action.

- **Fiducial systems (AprilTag/ArUco).** Established for pose estimation and robotics; here they are strictly candidates and spatial anchors inside a signed mapping, with authority excluded (P8, Annex XV).
- **Check-digit practice (ISO 7064, Damm, GS1 check digits).** Established error-detection art, adopted here as the Damm algorithm over an order-32 quasigroup applied to a variable-length, extensible identifier (§XVI.5.4).
- **Barcode/label generation suites and code-management platforms.** Established art generates and manages symbologies per format, commonly minting per-format links or per-campaign identities. The WR Code Generator differs in its coupling, not its rendering: one versioned entry record emits all carriers, emission itself writes the Carrier Binding Registry that runtimes later validate against, artifacts carry no secrets, and no emitted artifact constitutes a separate identity (§XVI.10).
- **Email graphic embedding.** Established practice renders sender-supplied images in mail clients. Under this annex received renderings are parse-only and never displayed; visuals are locally re-generated from validated identifiers (P12) — a discipline made practical by the same local rendering capability the generator defines.

The composite position — a single transport-agnostic identifier class with permanent, never-reassigned identifiers; a mandatory seven-layer resolution and consent chain uniform across all carriers; a universal human-readable baseline with augmentation-only machine carriers; a single generator whose emission act is registration in a runtime-validated Carrier Binding Registry; session-bound resolution of the same identifier into dynamic sub-entries without chain shortening; local re-rendering in place of display of received carrier artifacts; and a discovery layer under a non-tracking regime — is disclosed here as prior art in its combination.

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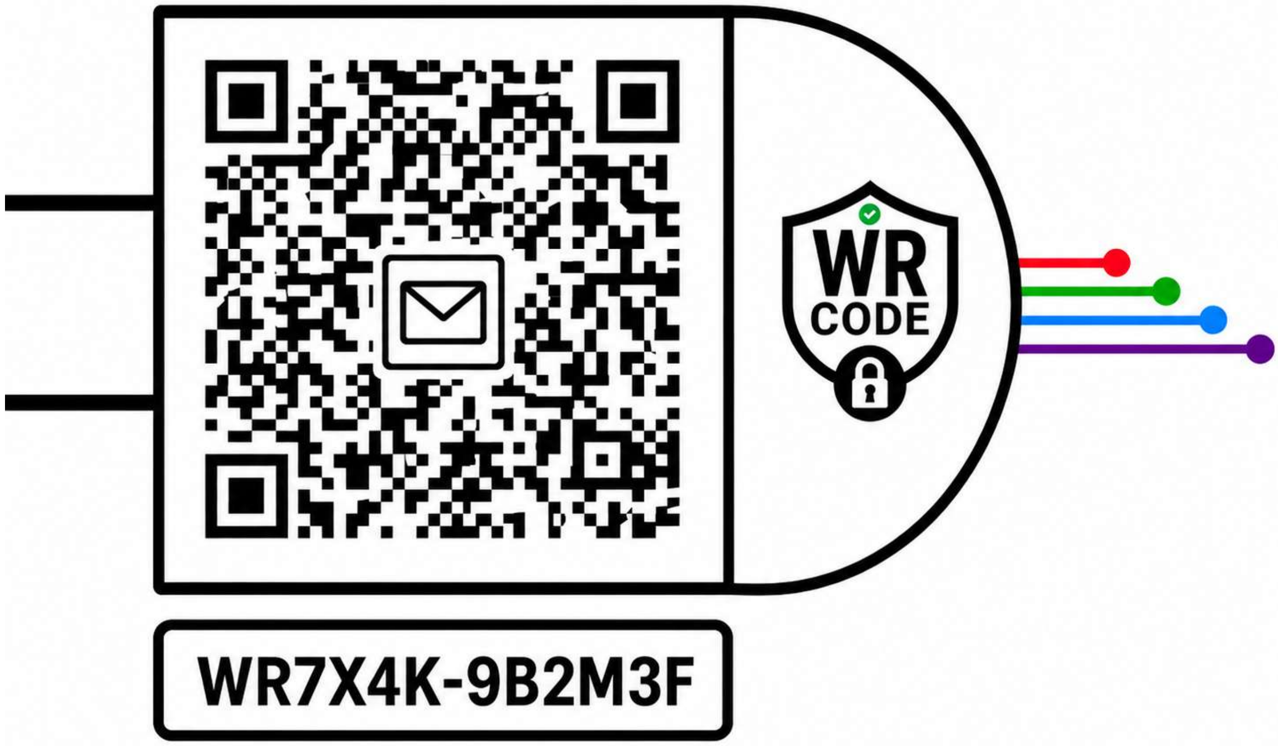
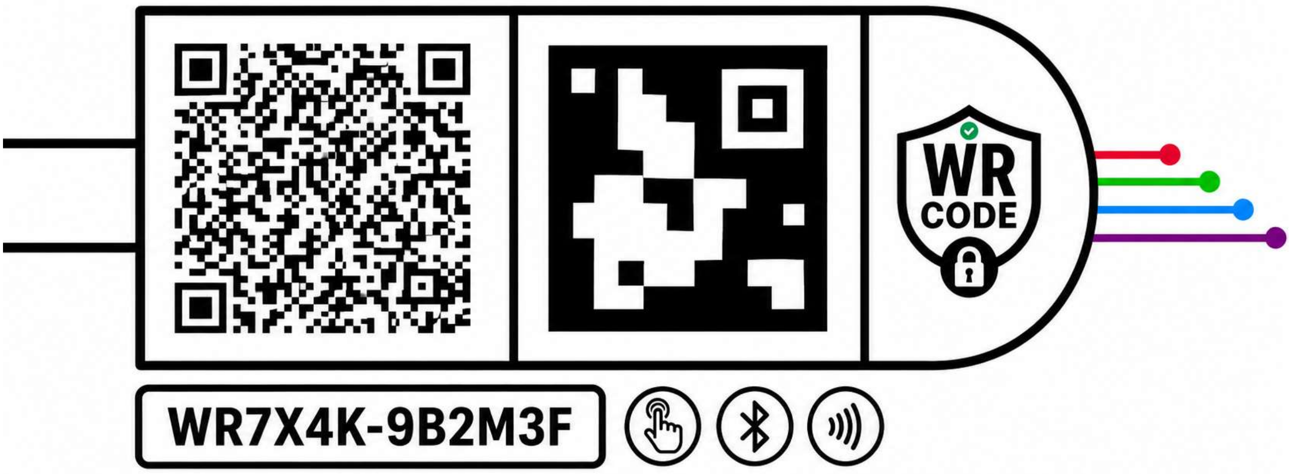
§XVI.18 Version History

- **v1.2 (5 August 2026).** Identifier scheme fixed normatively (§XVI.5.3–5.4): the Crockford Base32 alphabet (0–9, A–Z excluding I/L/O/U; case-insensitive with the Crockford input mapping) and the Damm check character over a totally anti-symmetric order-32 quasigroup replace the previously deferred registry profiles; the concrete quasigroup table and its test vectors remain registry-published material; the prior-work distinction on check-digit practice is updated accordingly. §XVI.13.1 restated as "Resolution contexts and entry granularity": a publisher MAY publish any number of context-specific entry codes under one Umbrella Handshake, each resolving its concrete context immediately without a session; Session-Bound Resolution is scoped to dynamically generated, non-pre-published sub-entries. §XVI.10.5 extended by the display-to-device handoff rule (locally re-rendered codes as a first-class cross-device capture surface — e.g., a smartphone capturing the inbox-displayed code and serving as voice input for the resolved context). §XVI.9.4 deployment examples marked as conceptual design patterns describing no implemented deployment. The open-items section of v1.1 (§XVI.18, E1–E10) is removed — a prior-art disclosure names no gaps — and its extension statements are folded into the governing sections as profile-admission and implementation-profile clauses; Version History is renumbered to §XVI.18. Crossreference corrections (§XVI.1.4 Floor Entry profile reference; §XVI.9.4 catalogue reference).

- **v1.1 (4 August 2026).** Full revision. New core principles P11 (identifier permanence) and P12 (local re-rendering). New §XVI.8 Entry Lifecycle and Identifier Stability: normative status model (active / inactive / revoked / superseded / compromised) with per-status runtime behavior, permanence-by-default with `expires_at` as the exception, once-only assignment of publisher parts and published local parts without quarantine, and the stable-identifier / rotatable-trust-material separation. New §XVI.10 WR Code Generator: one-record-many-renderings obligation, emission-is-registration coupling to the Carrier Binding Registry, Output Profiles (baseline-only through combined physical, print, email, and web packages), rendering requirement classes with registry-published parameter profiles, no-secrets rule, and the local re-rendering surface. New §XVI.13 Session-Bound Resolution and Dynamic Sub-Entries: context-free vs. session-attached resolution of the same identifier, Umbrella Handshake, session-scoped sub-entry identifiers, inherited publisher identity without inherited authority, and full chain discipline. §XVI.6.5 extended by the received-renderings-never-displayed rule and the forwarding/mailing-list note. §XVI.12 extended by Deployment Site definition and validation scoping (neighboring-entry rule). Carrier profiles: BLE layer-3 wording made normative (runtime-mediated only); capability-matrix column renamed to "Universal Baseline Carrier" with clarifying note; generic admission clause for further optical symbologies; new informative Carrier Selection Guidance (§XVI.9.4). §XVI.5.4 extended by the length-error analysis. Annex IX identifier-class boundary stated in §XVI.1.4. Security considerations extended: stale-referent capture, rendering divergence, enumeration and harvesting, resolution metadata privacy, repository availability. Prior-work section extended (generation suites, email graphic embedding) and composite position updated. Open items extended to E8–E10. Section renumbering: former §XVI.8–15 became §XVI.9, 11, 12, 14, 15, 16, 17, 18.
- **v1.0 (3 August 2026).** Initial version. Non-Identity Clause and canonical transport-agnostic definition; core principles P1–P10; seven-layer resolution model with layer discipline; 12-character Baseline Code (6+5+1) with extension rule, alphabet and check-character requirements; full-address vs. pointer payload classes; publisher determination profiles and email-carrier exception; verification-gated email capture flow with WR Connect affordance and same-surface manual entry; Universal Baseline Rule and co-presentation; carrier profiles and Carrier Capability Matrix; Bootstrap Payload with publisher-domain and central-resolver profiles; Carrier Binding Registry with baseline-first discipline and injection/cloning threat scoping; composite deployment catalogue; BLE discovery layer with normative privacy regime; offline operation; security considerations; prior-work distinctions; open items E1–E7.


WR
CODE
WR7X4K-9B2M3F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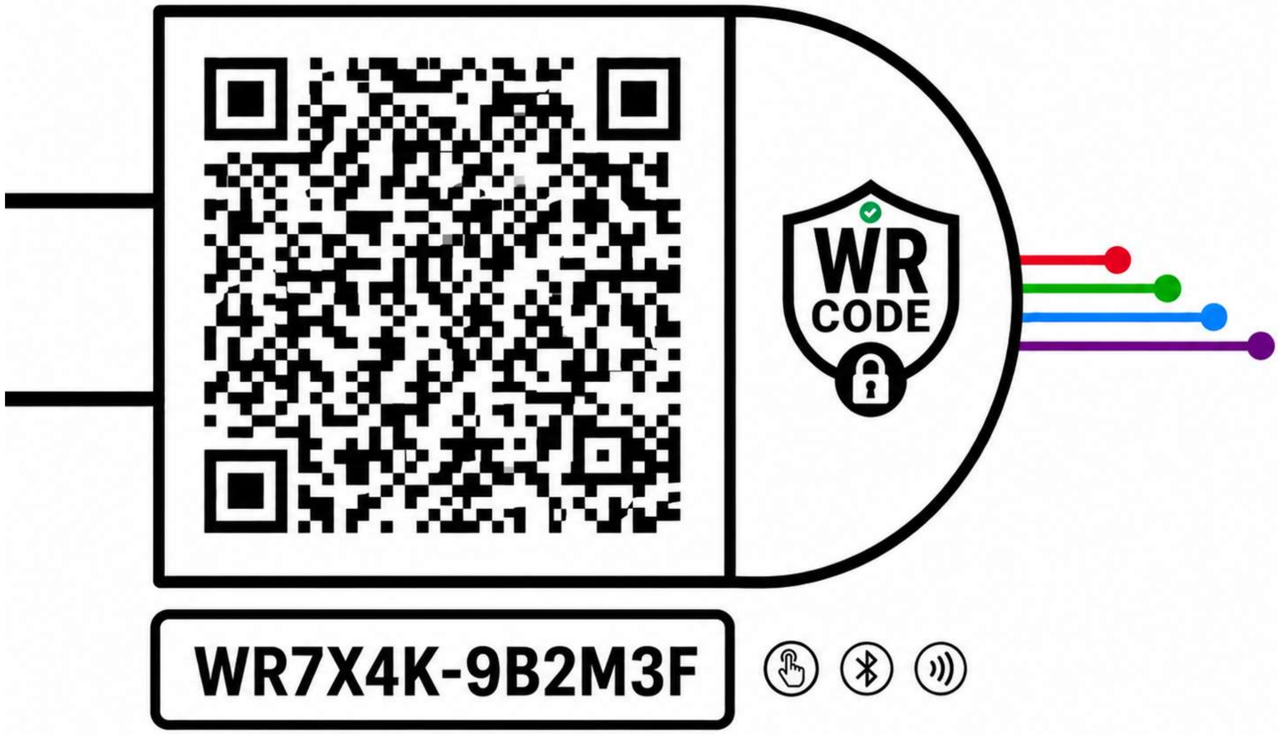
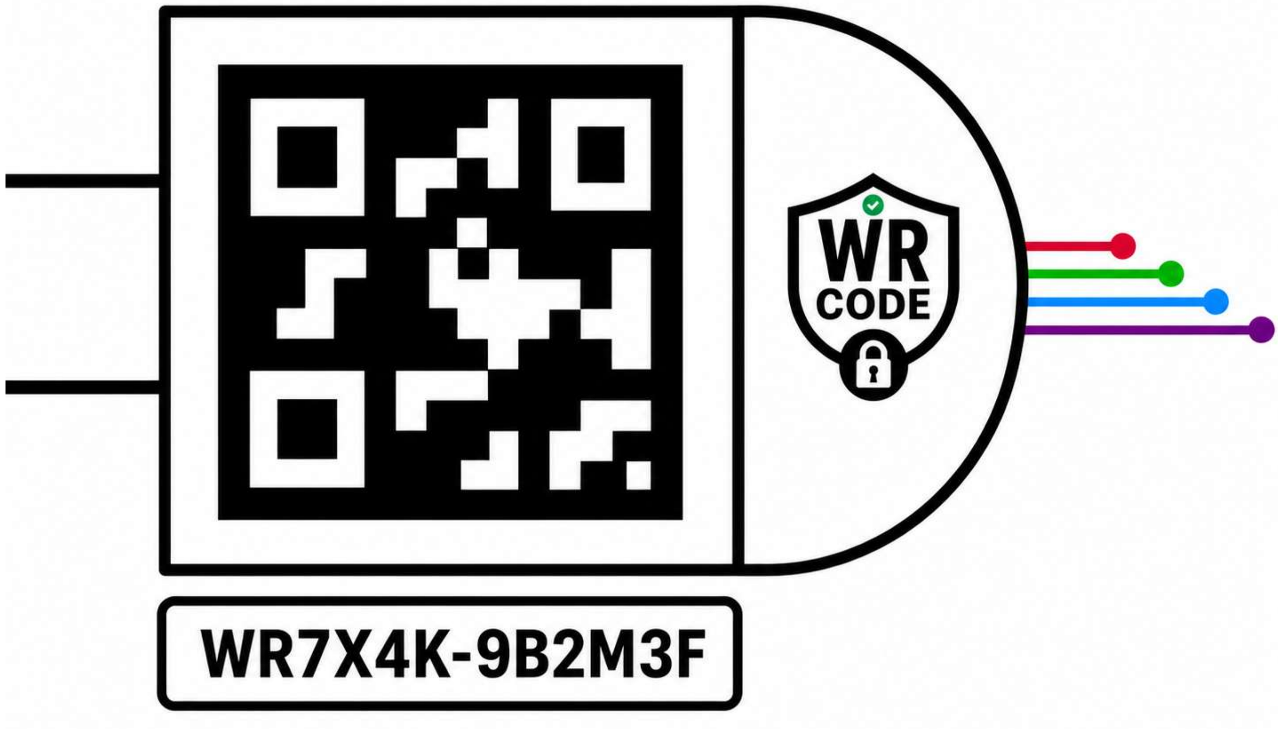

WR
CODE
WR7X4K-9B2M3F

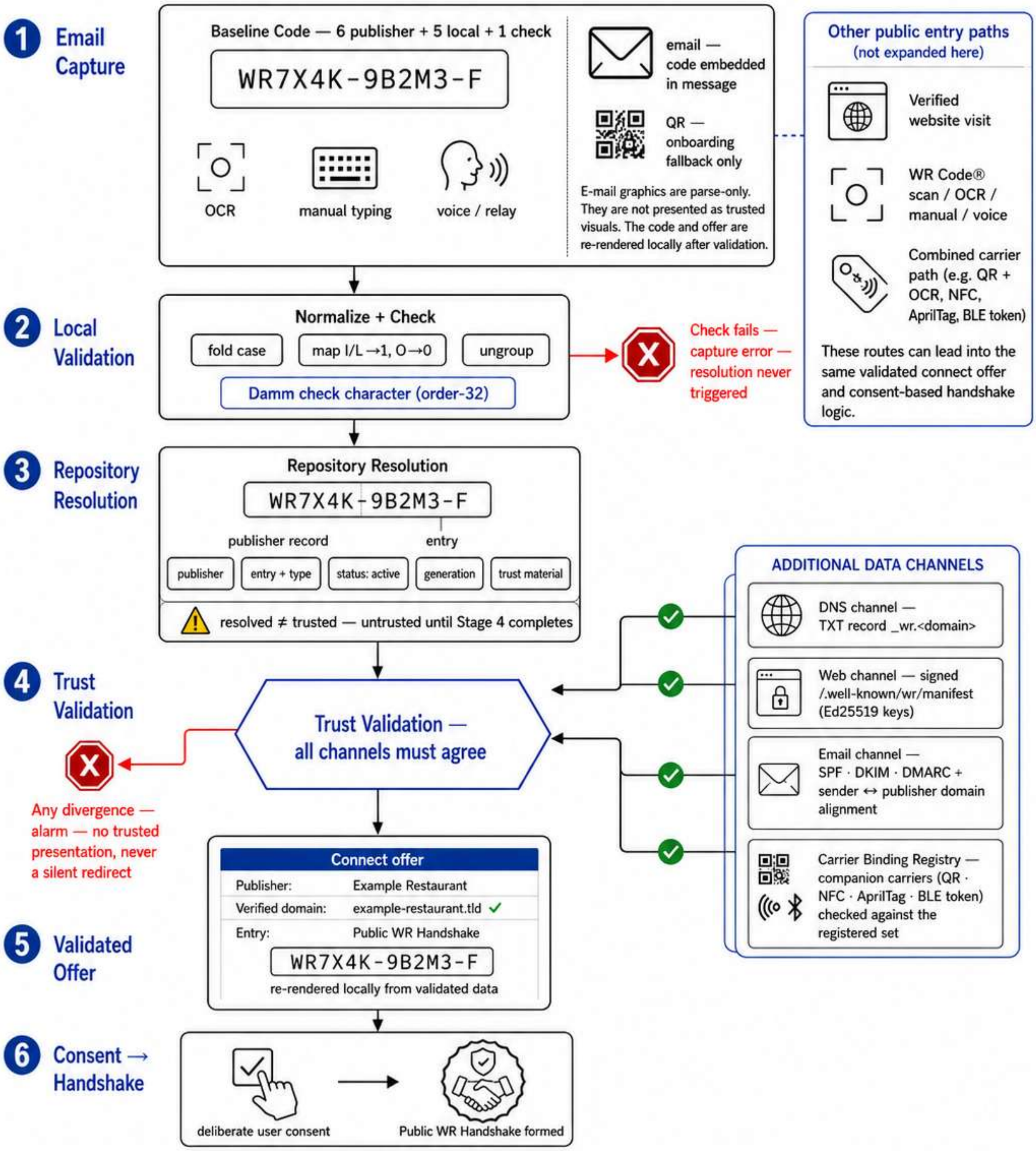

WR
CODE
WR7X4K-9B2M3F

WR Code® — Public Handshake via Email

Example path: e-mail-based verification. Other public entry routes also exist and converge into the same trust and consent model.

Scope of this figure

This figure explains the e-mail-based public handshake path. It does not represent all public entry paths. Other valid entry routes include visiting a verified website, scanning a WR Code® via OCR/manual/voice, and combined carrier paths such as QR + OCR, NFC, AprilTag, or BLE token.

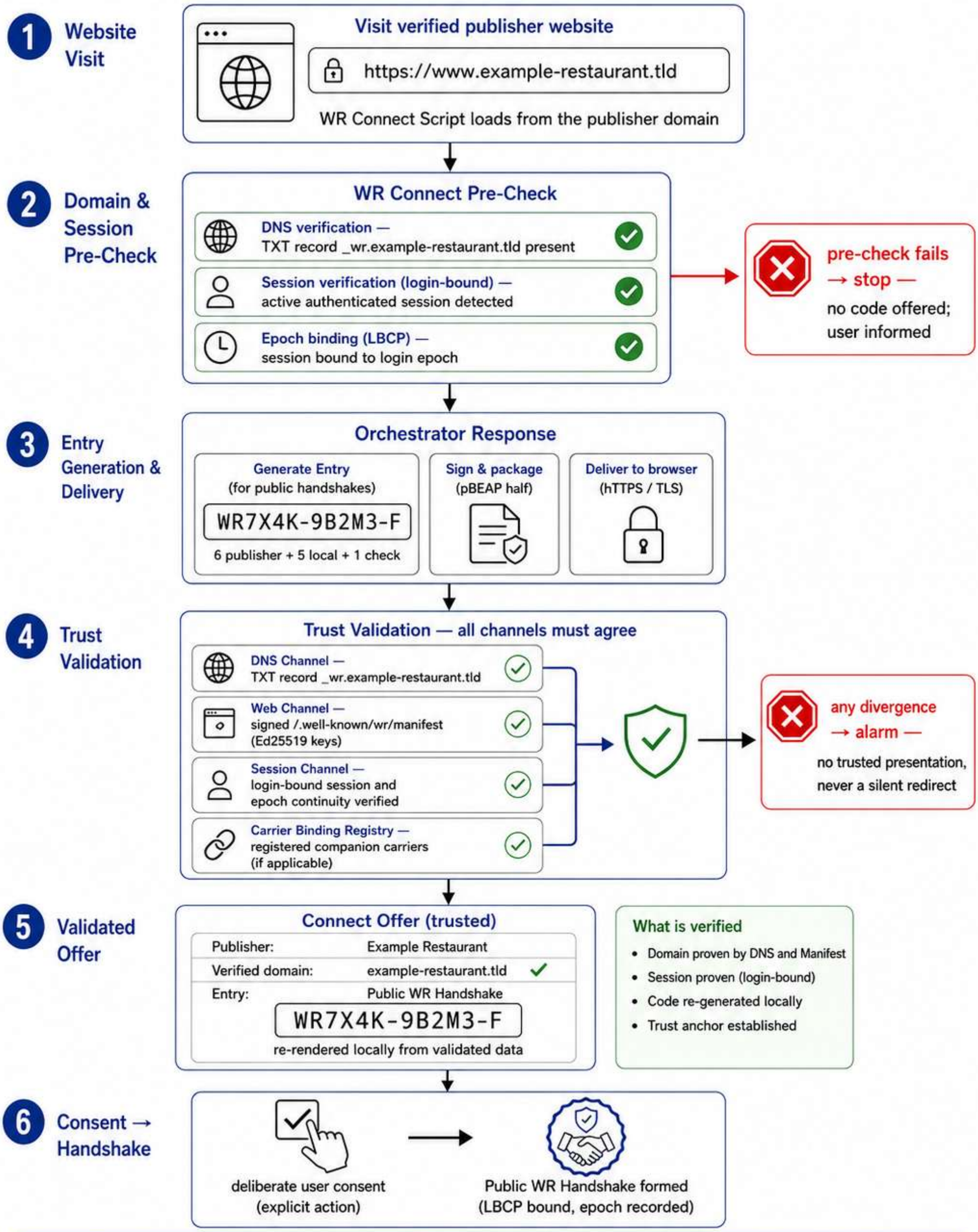


Recognition is never execution. This figure shows the e-mail path for a public handshake.
Other entry paths exist, but trust still arises only from resolution and validation.
— Annex XVI, Optirando™ Canon

Conceptual depiction. Example identifier only — no real code, product, or final visual design.

WR Code® — Public Handshake via Website Visit

Website path only — focused depiction of the verified web entry route.

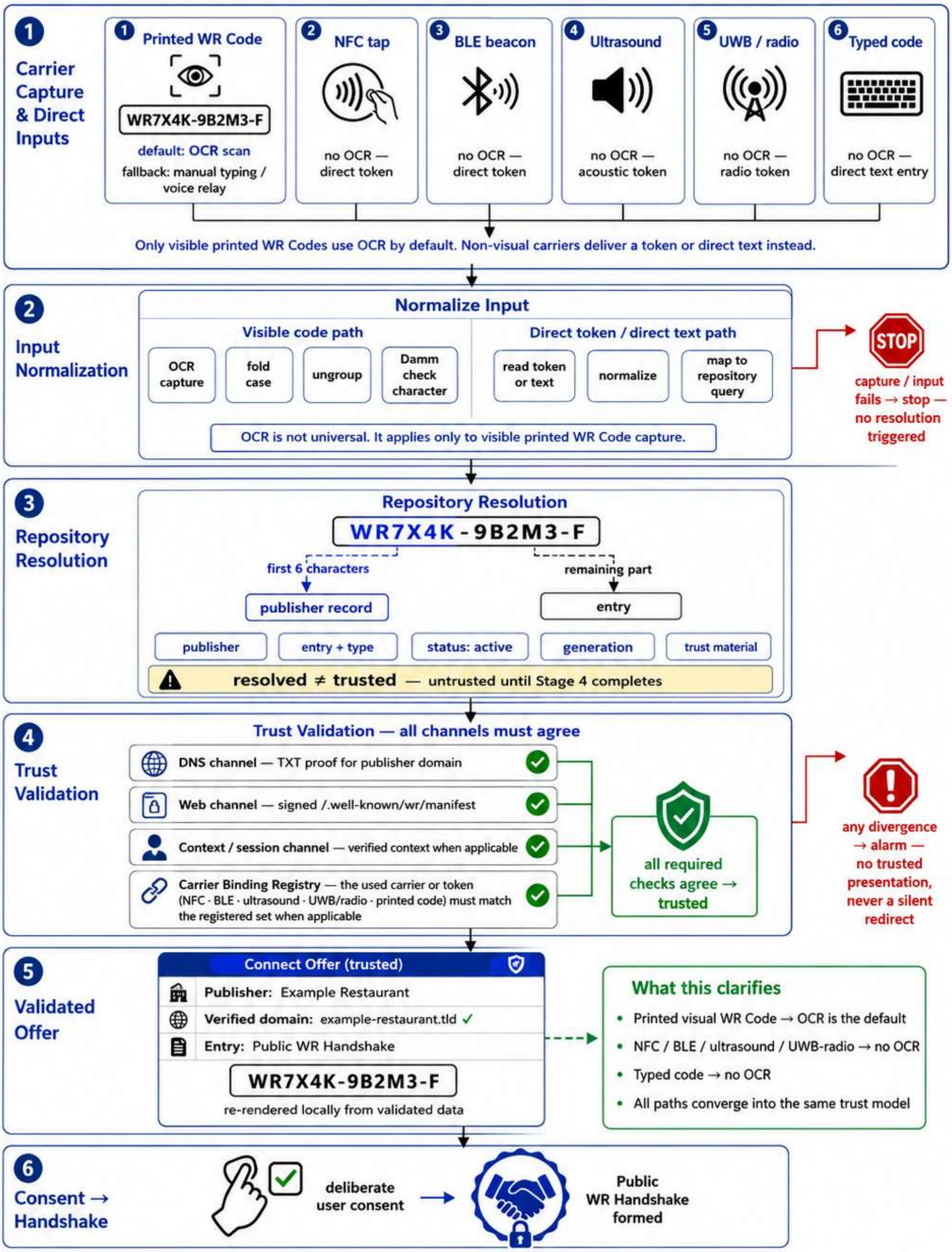


Recognition is never execution. Trust arises only from resolution and validation.
— Annex XVI, Optirando™ Canon

Conceptual depiction. Example identifier only — no real code, product, or final visual design.

WR Code® — Combined Carrier Paths

One identifier, multiple carriers — OCR is the default only for visible printed WR Codes.

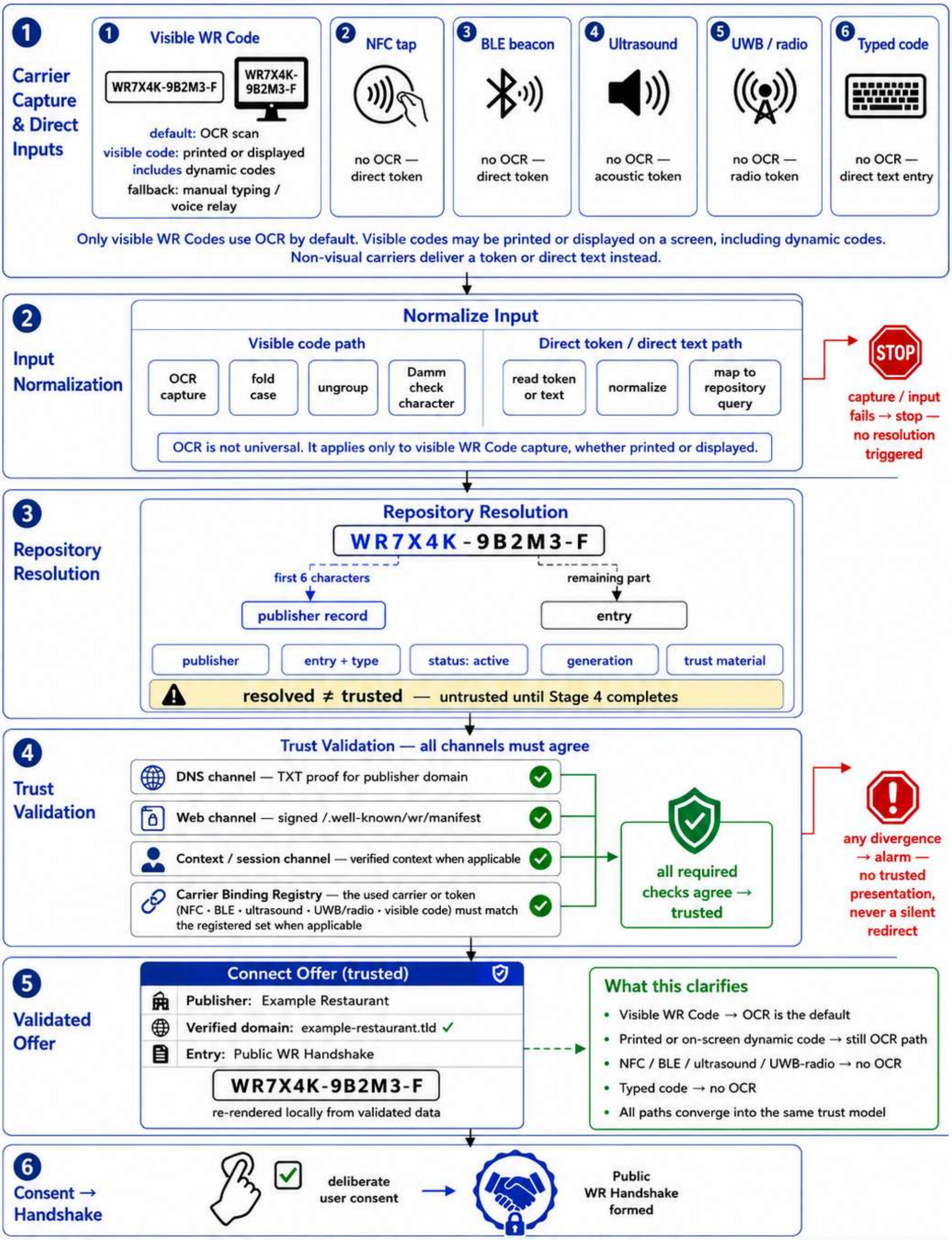


Recognition is never execution. Carriers transport identifiers or tokens — trust arises only from resolution and validation. — Annex XVI, Optirando™ Canon

Conceptual depiction. Example identifier only — no real code, product, or final visual design.

WR Code® — Combined Carrier Paths

One identifier, multiple carriers — OCR is the default only for visible WR Codes, including printed and on-screen dynamic codes.



Recognition is never execution. Carriers transport identifiers or tokens — trust arises only from resolution and validation. — Annex XVI, Optirando™ Canon

Conceptual depiction. Example identifier only — no real code, product, or final visual design.